A Power in and 6 V virtual ground

B Triangle carrier ~250 kHz (LM311 Schmitt + TL074 integrator)

H Indicator and bring-up

C Input stage (level -> buffer -> inverter)

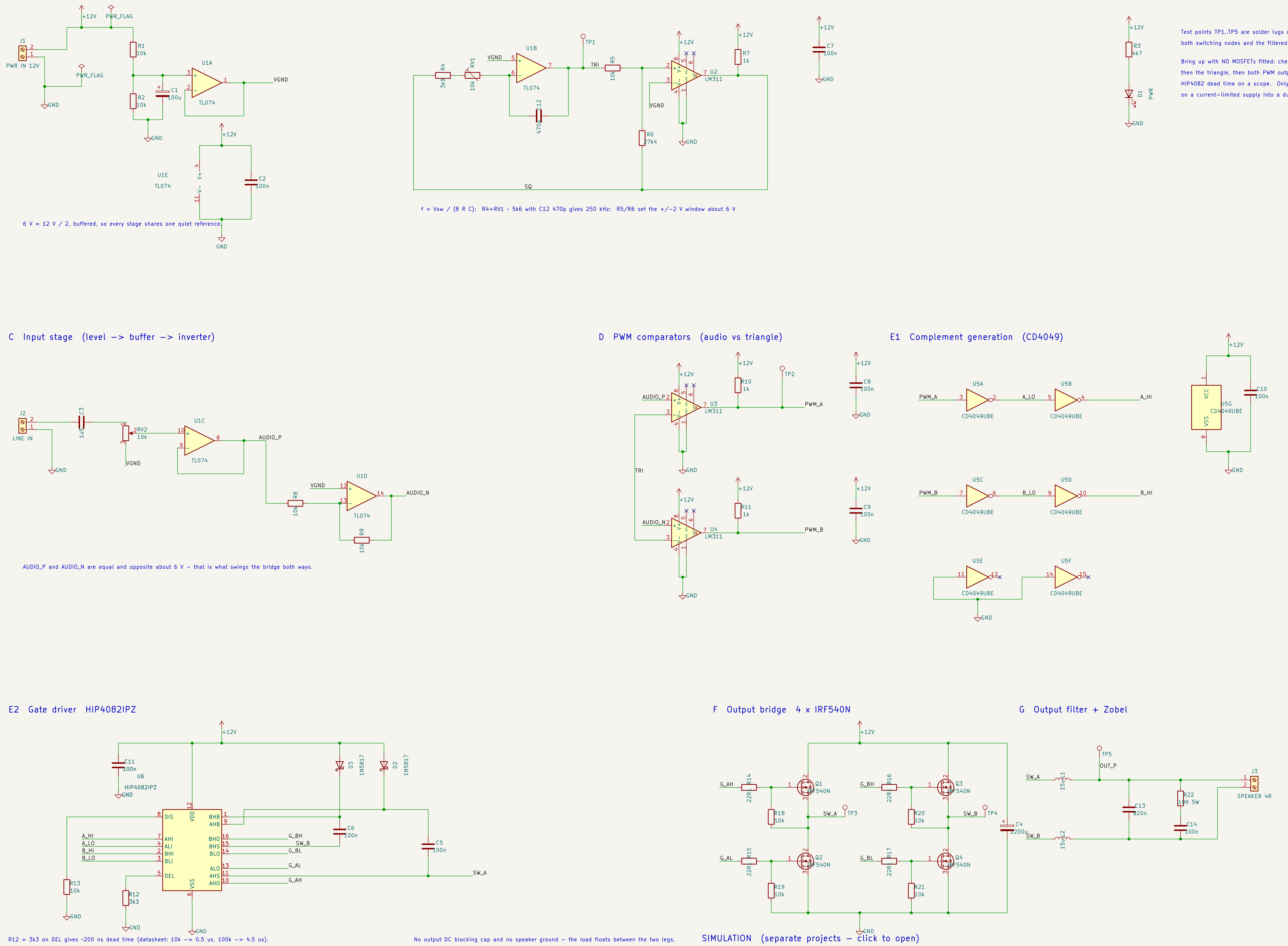
D PWM comparators (audio vs triangle)

E1 Complement generation (CD4049)

E2 Gate driver HIP4082IPZ

F Output bridge 4 x IRF540N

G Output filter + Zobel



6 V = 12 V / 2, buffered, so every stage shares one quiet reference.

f = Vsw / (8 R C): R4+RV1 ~ 5k6 with C12 470p gives 250 kHz; R5/R6 set the +/-2 V window about 6 V

Test points TP1..TP5 are solder lugs on TRI, PWM_A, both switching nodes and the filtered output.

Bring up with NO MOSFETs fitted: check 12 V and 6 V, then the triangle, then both PWM outputs, then the HIP4082 dead time on a scope. Only then fit Q1..Q4, on a current-limited supply into a dummy load.

AUDIO_P and AUDIO_N are equal and opposite about 6 V - that is what swings the bridge both ways.

R12 = 3k3 on DEL gives ~200 ns dead time (datasheet: 10k -> 0.5 us, 100k -> 4.5 us).

DIS is active high, so R13 holds it down.

No output DC blocking cap and no speaker ground - the load floats between the two legs.

L1/L2 are hand-wound on shop toroids: the stocked radial chokes saturate below the 3 A peak.

SIMULATION (separate projects - click to open)

A rails and the 6 V virtual ground -> sim/sim_a_vground.kicad_pro

B triangle carrier, ~250 kHz -> sim/sim_b_triangle.kicad_pro

C input stage, buffer and inverter -> sim/sim_c_input.kicad_pro

D PWM comparators -> sim/sim_d_pwm.kicad_pro

Sheet: /
File: classd.kicad_sch

Title: Class D amplifier - mono BTL, 12 V, 4 ohm

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 1/1